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Fixed-Orientation Sliding Blocks

Contact Geometry, Guide Wrench, and the Common Boundary

A sliding centrifugal weight can move outward while two surfaces determine its axial position. The fixed cup carries one surface; the movable sheave carries the other. Once a guide prevents relative rotation of the weight, those two contact constraints can determine its translation from sheave closure alone. The resulting motion supplies the same effective mass and shaft inertia as any other mechanism admitted by the common boundary [1].

That reduction does not make the block torque-free. Contact forces generally have different moment arms about its centre of mass (COM), and the block still rotates in space with the shaft. This note derives the contact motion first, then the inertia functions, complete ideal-guide demand, and the impulses needed to retain the contacts at a velocity reset.

The supported model is a CVTech-style sliding-block arrangement. The examples are synthetic profiles, not measured CVTech hardware. A particular cup, weight, insert, moving sheave, and anti-rocking guide still require their actual geometry and mass properties before this model can represent that assembly.

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1 Attachments, coordinates, and retained motion

Use the right-handed local basis $(\boldsymbol{e}_r, \boldsymbol{e}_t, \boldsymbol{e}_z)$, with $\boldsymbol{e}_z$ along the shaft and $\boldsymbol{e}_t = \boldsymbol{e}_z \times \boldsymbol{e}_r$. The local sheave coordinate x is positive in closure; the moving contact track translates by $+x\boldsymbol{e}_z$. The lower track is axially fixed in the shaft frame. Both tracks and the orientation guide co-rotate with the shaft at angular speed ω and acceleration α. The origin of x is arbitrary: positive denotes its direction, not a restriction $x \geq 0$. In particular, CINDER's secondary opens into negative local coordinates during upshift. The supplied profiles and compiled interval must cover that signed travel.

Let $(R(x), Z(x))$ locate a datum fixed in the block. The block does not rotate relative to these local axes. Its COM offset from the datum is the fixed vector (c_r, c_t, c_z), so the COM coordinates are

$$r = R + c_r, \qquad y = c_t, \qquad z = Z + c_z. \tag{1.1}$$

Separating this datum from the COM is essential for tuning inserts: changing an insert can move the COM without moving either contacting face.

The complete identical set has mass M and contains n blocks. One block has mass M/n. Store the centroidal inertia tensor per unit mass as $\boldsymbol{K}$, expressed in (r, t, z) axes. Thus $\boldsymbol{I}_{C,\mathrm{set}} = M\boldsymbol{K}$ is the sum of corresponding local centroidal tensors, and $\boldsymbol{I}_{C,1} = M\boldsymbol{K}/n$ belongs to one block. Off-diagonal entries are the entries of the inertia tensor itself: for example, $K_{rz} = -\int u_r u_z \, \mathrm{d}m/m$. They are not unsigned geometric products of inertia.

The mass set is constant. The cup and rigid sheave masses remain in the host pulley accounting; they are not included in M. Springs and belt loads are separate elements. The contacts here are frictionless and unilateral. The initial guide is an explicit ideal constraint: it supplies a tangential force through the COM and an arbitrary holding couple, with no frictional loss. Physical guide capacities are unknown, rather than inferred from the outline.

2 Two finite profiles determine the body translation

Each contact face is a finite graph in a meridional section. Write the lower and upper block faces as $z = b_\ell(u)$ and $z = b_u(u)$, where u is radial position relative to the body datum. The two faces share a declared interval $[u_a, u_b]$, and $b_u > b_\ell$ throughout it. The fixed and moving tracks are $z = f_\ell(\rho)$ and $z = f_u(\rho) + x$, respectively, on finite radial domains. The whole block footprint must remain within both modelled track domains.

At a trial datum pose (R, Z), define the signed axial gaps

$$g_\ell(u; R, Z) = Z + b_\ell(u) - f_\ell(R + u), \tag{2.1}$$

$$g_u(u; R, Z, x) = f_u(R + u) + x - Z - b_u(u). \tag{2.2}$$

A nonnegative gap means separation without penetration. A retained contact requires the minimum gap on that face to be zero. This compares the full finite face, rather than selecting a tangent while ignoring another part of the same face that might penetrate the track.

The contact equations above do not prescribe a circular cup or a particular block outline. CINDER 1.2.1 accepts user-defined twice continuously differentiable piecewise polynomial graphs, alongside exact lines, quadratics and circular-arc graphs. On a radial interval $[a_j, a_{j+1}]$, a polynomial profile

is stored as

$$f(\rho) = f_0 + \sum_{k=0}^{d_j} c_{jk} t^k, \qquad t = \frac{\rho - a_j}{a_{j+1} - a_j}. \tag{2.3}$$

The coefficients are axial lengths. Normalizing the independent coordinate avoids powers of very small metre-valued intervals in the root calculation. Seams agree through the second derivative. Point input explicitly interpolates a cubic spline, or a quintic when requested; it does not silently fit or smooth measurement noise. Curves and their actual derivatives, rather than a display mesh, determine contact.

These remain meridional graphs. Overhangs, independent rocking and three-dimensional saddle contact require a different contact construction. That is a scope of this translation provider, not a restriction of the common material-motion boundary. The block's full mass distribution remains separate from its selected contacting faces.

2.1 Finding the whole contact set

The minimum gap can occur at an endpoint, at a profile seam, at any stationary point, or throughout a constant-gap interval. The implementation enumerates all of these candidates over the finite face. For polynomial pieces, the gap is polynomial and its derivative roots supply all interior candidates, including repeated stationary roots. Derivative roots partition the normalized interval into monotone pieces for bracketed root isolation. Curvature may change sign and there may be several local minima; neither condition is a reason to select only one guessed tangent.

With a circular graph, stationarity involves a square root. Squaring produces a polynomial candidate equation. Every candidate is then checked against the original unsquared slope equation to remove extraneous roots. The opposite branches of a circle are not mistaken for a constant gap merely because their squared derivative expressions agree. These constructions are evaluated in floating-point arithmetic with declared numerical resolutions.

Retain all candidates attaining the minimum to that resolution. A pair (u, u) denotes a point and $[a, b]$ a complete flat contact interval. The representative coordinate used to draw a normal is not a pressure centroid. This distinction will matter for the guide moment even when the clamping resultant is determined.

2.2 Establishing a regular configuration branch

To reduce the two equations, put

$$A(R) = \max_u [f_\ell(R + u) - b_\ell(u)], \qquad C(R) = \min_u [f_u(R + u) - b_u(u)]. \tag{2.4}$$

The lower contact gives $Z = A(R)$ and the upper contact gives

$$x = A(R) - C(R). \tag{2.5}$$

Disjoint bounds on the slopes of the active tracks establish that the derivative of this function is nonzero. A bracketed scalar solve then gives the unique R on a selected regular component of the radial search window. If the window contains a fold, subdivision isolates monotone components. The component must cover the full requested axial travel; an optional `reference_radius` selects the assembly when several components fit. Track-wide bounds are tried first. If they overlap, a

contact enclosure excludes regions of the face which cannot attain the minimum gap, and the radial window is subdivided. To see the exclusion, let a trial face point give a uniform upper bound on the minimum gap over a radial interval. Any face subinterval whose lower gap bound exceeds it cannot contain contact. Profile extrema bound both values and derivatives throughout each interval.

Monotone closure alone does not establish smooth motion: the identity of the minimum can switch between separated parts of a curved track. On the reachable radial path, additional local bounds establish an isolated regular minimum or a retained endpoint. Affine tracks admit an important further case: the gap separates into a function of u and an affine function of R. Its minimizing contact set is fixed even if it contains several points or a whole face, so its envelope remains smooth. Section 6 treats the associated loads.

A numerical enclosure failure is not proof that the proposed hardware is impossible. The implementation reports the unresolved interval and contact. It does not hide a corner by blending two body-motion curves. Contact-set cardinality may change while the body motion remains smooth. For example, a flat block face can span several coplanar pads. Subtracting its slope from the track leaves a constant extremum on each pad. If at least one extremal pad intersects the footprint at every R, the envelope is affine, even as its contact intervals split or disappear.

A concave gap has only two endpoint candidates. Their polynomial crossings are isolated with 80-digit arithmetic and their winning branches compared on either side. A switch with matching value and first two derivatives retains a C^2 body motion and becomes an interpolation boundary. A switch with incompatible derivatives requires a nonsmooth event model and is reported as outside this smooth provider, rather than invalid hardware. Other unresolved competing minima remain numerical failures. General profile extrema still use floating-point stationary-root calculations; interpolation diagnostics are separate sampled checks.

Endpoint contact is allowed only when explicitly requested. It is an ideal point contact at a finite face endpoint with the smooth track's normal. A rounded nose of finite radius must be represented by its actual geometry if its change to the contact path matters. Neither an unknown corner radius nor a pressure centroid is fitted automatically.

3 Motion ratios from the same contact constraints

Let u_ℓ and u_u be the active face coordinates and let $s_i = f_i'(R + u_i)$ be the corresponding track slopes. At an interior tangent, $b_i'(u_i) = f_i'(R + u_i)$. The derivative of the minimum gap with respect to its minimizer vanishes; at a retained endpoint, u_i is constant. In either case, differentiating the two closed gaps gives

$$Z' = s_\ell R', \qquad Z' = 1 + s_u R'. \tag{3.1}$$

Consequently,

$$D = s_\ell - s_u, \qquad R' = \frac{1}{D}, \qquad Z' = \frac{s_\ell}{D}. \tag{3.2}$$

The constraint Jacobian has rows $(-s_\ell, 1)$ and $(s_u, -1)$. As D approaches zero the geometry ceases to determine a regular translation. The implementation compares slope separation with arithmetic resolution and checks the resulting motion accuracy. It does not impose the earlier fixed condition-number cutoff or clip a large motion ratio.

Curvature changes the active track slope as the block moves. At an interior contact, differentiation of the tangency condition with respect to R yields

$$u_{i,R} = \frac{f_i''}{b_i'' - f_i''}, \qquad s_{i,R} = f_i''(1 + u_{i,R}). \tag{3.3}$$

At a retained endpoint $u_{i,R} = 0$, including a pose where both endpoints touch. When several support candidates share their value and slope, the lower maximum envelope takes the greatest candidate second derivative; the upper minimum takes the least. An isolated touching candidate need not remain active on either side. All profile derivatives in this expression are evaluated at the active material coordinates. Thus the second motion ratios are

$$R'' = -(s_{\ell,R} - s_{u,R})(R')^3, \qquad Z'' = s_\ell R'' + s_{\ell,R}(R')^2. \tag{3.4}$$

The fixed COM offset does not affect these derivatives: $r' = R'$, $z' = Z'$, and likewise for second derivatives.

3.1 Mass geometry is independent of contact selection

The contact faces determine compatible motion; the retained material determines mass, COM and inertia. A uniform extruded polygon can be reduced using its area moments. A smooth body bounded by two profiles can instead be integrated directly, without using the tessellated drawing. Both constructions return the COM and full centroidal tensor per unit mass in the body datum; the supplied total group mass fixes the uniform density. Inserts or nonuniform material require their actual moments. Additional cup or sheave backing is CVT material and never enters this weight-mass reduction.

4 Mass motion supplies the four quantities

Because orientation is fixed relative to the shaft frame, every material point has the same relative translation velocity $(r', 0, z')\dot{x}$. Its spin about the COM still contributes to shaft-rotation energy. Integrating the material velocity therefore gives

$$T_f = \tfrac{1}{2}M[(r')^2 + (z')^2]\dot{x}^2 - Mc_t r'\omega\dot{x} + \tfrac{1}{2}[M(r^2 + c_t^2) + MK_{zz}]\omega^2. \tag{4.1}$$

The mixed coefficient is $B_f = -Mc_t r'$. Fixed orientation alone does not remove it. The initial implementation requires $c_t = 0$; it does not infer cancellation from equally spaced blocks. With that condition established,

$$\boxed{\begin{aligned} M_f &= M[(r')^2 + (z')^2], \quad M_f' = 2M(r'r'' + z'z''), \\ J_f &= Mr^2 + MK_{zz}, \qquad J_f' = 2Mrr'. \end{aligned}} \tag{4.2}$$

These are total-set quantities. The count n does not multiply them again. The tensor must be finite and symmetric, with nonnegative principal inertias satisfying the triangle inequalities. These checks establish a possible inertia tensor; they do not establish that supplied mass properties actually belong to the specified outline and inserts.

The common CINDER element [1] consumes equation (4.2) and supplies

$$F_x = \tfrac{1}{2}J_f'\omega^2 - M_f\ddot{x} - \tfrac{1}{2}M_f'\dot{x}^2, \qquad \tau = -J_f\alpha - J_f'\dot{x}\omega. \tag{4.3}$$

There is one implementation of these relations in `CentrifugalInertiaForce`. The sliding-block provider adds the geometry and contact policy; it does not carry another transient force law. The same M_f and J_f enter kinetic energy and event projection.

5 Contact reactions and the torque on each block

The four maps describe work and kinetic energy, but their reduction has eliminated the individual constraint reactions. Recover those reactions from the block's free body before deciding whether the contact branch is admissible. The contact normals on the block, written as radial–axial pairs, are

$$\boldsymbol{n}_\ell = \frac{(-s_\ell, 1)}{\sqrt{1+s_\ell^2}}, \qquad \boldsymbol{n}_u = \frac{(s_u, -1)}{\sqrt{1+s_u^2}}. \tag{5.1}$$

With compression positive, the total corresponding normals satisfy

$$[\boldsymbol{n}_\ell \ \boldsymbol{n}_u] \begin{bmatrix} N_\ell \\ N_u \end{bmatrix} = M \begin{bmatrix} r'\ddot{x} + r''\dot{x}^2 - r\omega^2 \\ z'\ddot{x} + z''\dot{x}^2 \end{bmatrix}. \tag{5.2}$$

The tangential acceleration is separate. The ideal guide supplies

$$G_t = M(r\alpha + 2\omega r'\dot{x}). \tag{5.3}$$

The $2\omega r'\dot{x}$ term is the tangential acceleration associated with radial motion in the rotating frame. Neither contact normal in this model has a tangential component.

The guide must also prevent relative rocking. Let $\boldsymbol{a}_i$ run from the COM to contact i, let $\boldsymbol{N}_i = N_i(n_{i,r}, 0, n_{i,z})$, and use the full centroidal tensor. The required holding couple is

$$\boxed{\boldsymbol{G}_C = M\boldsymbol{K}\,(\alpha\boldsymbol{e}_z) + (\omega\boldsymbol{e}_z) \times [M\boldsymbol{K}(\omega\boldsymbol{e}_z)] - \sum_{i=\ell,u} \boldsymbol{a}_i \times \boldsymbol{N}_i.} \tag{5.4}$$

The spin terms have components

$$M(K_{rz}\alpha - K_{tz}\omega^2, \quad K_{tz}\alpha + K_{rz}\omega^2, \quad K_{zz}\alpha). \tag{5.5}$$

Even at constant shaft speed, a nonzero product of inertia can require a transverse couple. Even with zero products, the two contact moments need not cancel. A single shaft-axis gyration radius cannot determine this full demand.

For the straight synthetic case at $x = 7.5\,\mathrm{mm}$, $r = 57\,\mathrm{mm}$, $M = 0.75\,\mathrm{kg}$ and $\omega = 350\,\mathrm{rad\,s^{-1}}$, steady contact gives $F_x = 2094.75\,\mathrm{N}$. The contact-only holding couple has magnitude $51.0595\,\mathrm{N\,m}$ in corresponding local axes, or $17.0198\,\mathrm{N\,m}$ per block for $n = 3$. Figure 1 shows why the lines of action do not remove this moment.

All normals and guide components above are sums of corresponding local values; divide by n for one pocket. They are not a global transverse-vector sum around the pulley. Symmetric cancellation at assembly level does not unload each pocket. Circumferential sidewall forces alone also cannot resist a moment about the tangential axis: an anti-rocking feature must constrain that motion. A specified real guide may transmit radial and axial forces too, in which case equation (5.2) must be replaced by the complete guide/contact equilibrium.

For the present ideal guide, the moving track delivers $F_x = -N_u n_{u,z}$. This agrees with equation (4.3) for the exact constraint motion. The runtime contact policy uses both N_ℓ and N_u at the actual solved accelerations. It rejects resolved tension. Raw reactions remain available; a numerical-resolution check described below distinguishes unresolved near-zero values from a required tensile contact.

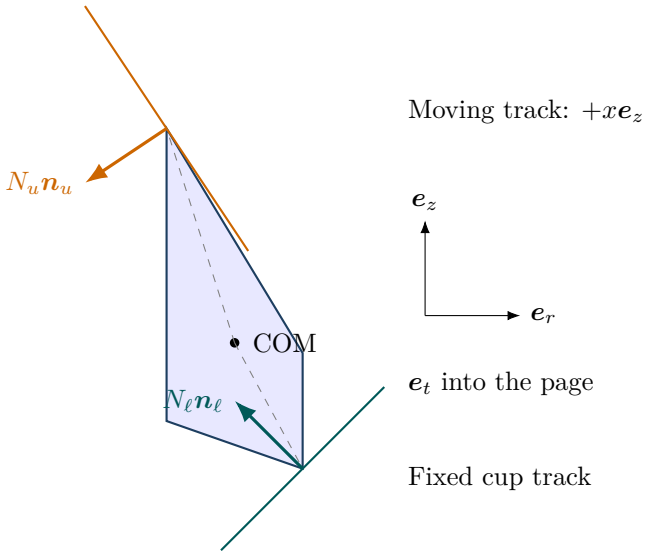


Figure 1: Local section of the straight synthetic block at the stated pose. The contact arms are measured from the COM. Shaft-induced centripetal acceleration is supplied by the resultant radial force; the guide couple must balance the contact moment as well as any centroidal spin demand. The track segments indicate contact surfaces, not complete cup or sheave solids.

6 Clamping with an undetermined pressure location

Two matching flat faces need not determine a single pressure application point. Nevertheless, the flat moving track has one normal direction and supplies one closed-gap equation. For $b_u(u) = b_0 + s_u u$ and $f_u(R + u) = f_0 + s_u(R + u)$, that equation is

$$Z = f_0 - b_0 + s_u R + x. \tag{6.1}$$

The coordinate u has cancelled. Combined with the fixed-cup envelope $Z = A(R)$, it can determine a unique regular translation. Its derivatives and the four inertia quantities follow as before. The two common normal directions then determine the total forces in equation (5.2). Thus unknown pressure location does not, by itself, make clamping undetermined.

A nonconvex block may likewise touch an affine track at two separated points. Both normals have the same direction, and their nonnegative pressures add to one force resultant. The pressure centroid can lie in the convex hull of the contact set, even if the material between the two points is not in contact. The API retains the actual separate points instead of drawing this convex hull as a physical contact patch.

The remaining ambiguity concerns moment. Let $\mathcal{U}_i$ be the contact set on face i and put

$$\boldsymbol{q}_i(u) = -\boldsymbol{a}_i(u) \times [N_i\boldsymbol{n}_i], \qquad \boldsymbol{G}_0 = M\boldsymbol{K}(\alpha\boldsymbol{e}_z) + (\omega\boldsymbol{e}_z) \times [M\boldsymbol{K}(\omega\boldsymbol{e}_z)]. \tag{6.2}$$

For each component k, the ideal-guide demand is bounded by

$$G_{0,k} + \sum_i \min_{u\in\mathcal{U}_i} q_{i,k}(u) \;\le G_{C,k} \le\; G_{0,k} + \sum_i \max_{u\in\mathcal{U}_i} q_{i,k}(u). \tag{6.3}$$

For the admitted common-normal meridional contacts, only the tangential component has pressure-location uncertainty. On a flat interval its moment is affine in u, so the endpoints determine its

extrema; isolated contact points are evaluated individually. These bounds include the full centroidal spin demand. A unique contact point gives coincident bounds and recovers equation (5.4). When these bounds differ, the single guide moment is marked undetermined. Coincident bounds still give a unique moment, including the unloaded case. The same construction applies to impulses using P_i and the centroidal angular-momentum jump.

As a closed-form check, take a fixed affine cup of slope 0.8, a matching moving face of slope -0.5, and a lower block face

$$b_\ell(u) = 0.8u + c(u^2 - a^2)^2, \quad a = 0.004\,\mathrm{m}, \quad c = 10^6\,\mathrm{m}^{-3}. \tag{6.4}$$

The lower face contacts at the two distinct points $u = \pm a$. With the open datum $R_0 = 0.05\,\mathrm{m}$, $Z_0 = 0$, and suitable upper-face thickness, the two gap constraints give

$$R = R_0 + \frac{x}{1.3}, \qquad Z = \frac{0.8x}{1.3}. \tag{6.5}$$

For zero COM offset and held closure, the mechanism's closing contribution is $F_x = Mr\omega^2/1.3$. Springs and belt loads enter the host balance separately. It is unique although the guide moment has a finite pressure-dependent range. This example tests multiple global minima and flat face contact without using the compiled motion as its expected solution.

7 Retaining the contacts through a velocity reset

An instantaneous reset changes velocities at one fixed configuration. Smooth centrifugal, Coriolis, spring, and applied-torque terms do not supply finite impulses over that zero-duration interval. If the same two contacts and guide remain active, their required impulses follow from the velocity jumps:

$$[\boldsymbol{n}_\ell\ \boldsymbol{n}_u] \begin{bmatrix} P_\ell \\ P_u \end{bmatrix} = M \begin{bmatrix} r' \\ z' \end{bmatrix} \Delta\dot{x}, \tag{7.1}$$

$$P_t = Mr\Delta\omega, \tag{7.2}$$

$$\boldsymbol{L}_C = M\boldsymbol{K}(\Delta\omega \boldsymbol{e}_z) - \sum_i \boldsymbol{a}_i \times \boldsymbol{P}_i. \tag{7.3}$$

The normal impulses must be compressive. The ideal guide is permitted to carry its computed force impulse and couple impulse, but its real capacity is still unknown. A positive running normal force does not establish the sign of these impulses.

For the straight case, arresting $\dot{x} = +0.1\,\mathrm{m\,s^{-1}}$ with no shaft-speed jump requires

$$(P_\ell, P_u) = (-0.00848528,\ 0.04326662)\,\mathrm{N\,s}. \tag{7.4}$$

The lower contact would have to pull. The stop therefore cannot retain this branch, despite admissible pre-impact running forces. CINDER exposes the normal-impulse margins on candidate projections and rejects an inadmissible selected transition. Resolving what happens after release needs a different contact topology or an additional dynamic coordinate; this version does not invent that motion.

8 Compilation, inputs, and limits in CINDER 1.2.1

`SlidingBlockGeometry` stores the four profiles, radial branch window, local axial interval, and explicit endpoint-contact policy. `SlidingBlockMass` stores total mass, count, COM offset, and the full centroidal tensor per mass. `SlidingBlockSpec` combines these with the required `ideal_fixed_orientation` guide declaration. The saved component kind is `sliding_block_flyweight`. Distances are metres, mass is kilograms, profile curvature is inverse metres, and tensor entries are square metres. The JSON examples in the package example folder provide complete component records, including a noncircular cup and a nonconvex multi-contact weight; they are not full CVT assemblies.

The compiler solves radial closure before integration and constructs a common C^2 piecewise quintic Hermite representation of R, Z. Both inertia functions and their derivatives use this same material motion. Affine track pairs give an exactly linear motion representation, avoiding artificial curvature from endpoint rounding over a very short axial interval.

Contact locations are separate geometric data. Stable contact sets retain cached endpoint paths. A changing or insufficiently smooth contact path instead uses a global finite-face query at the compiled body pose when reactions, impulses or a preview need contact locations. This query can perform polynomial root searches, but does not re-solve radial closure. Inertia-map evaluation always remains free of geometry solves. The compilation report identifies the contact-query strategy.

Starting from eight intervals and the identified smooth transition boundaries, the compiler tests three interior points per cell and subdivides failed cells. Its body-motion tolerances are $2 \times 10^{-10}\,\mathrm{m}$ for positions, 2×10^{-7} for first derivatives, and $2 \times 10^{-4}\,\mathrm{m}^{-1}$ for second derivatives. Cached contact paths have a separate sampled position tolerance of $2 \times 10^{-10}\,\mathrm{m}$. The report records the largest observed errors, the selected radial component, and observed contact counts. These are sampled interpolation diagnostics, not rigorous continuum bounds. Contact-count changes do not reject a regular body motion. Conservative regularity enclosures can still fail on valid inputs; this construction does not establish completeness for all graphs.

The cache is keyed by immutable geometry, so mass changes reuse the motion. Positive COM radius is required over reachable travel. Queries outside the compiled axial interval raise an error without extrapolation.

One sliding-block element represents the entire distributed group on a pulley. Python construction and saved-document validation reject duplicate groups; change the total mass instead. If an insert also changes the distribution, update the COM and tensor rather than scaling mass alone.

8.1 Numerical resolution at zero contact load

A mathematically zero reaction can evaluate slightly below zero when large inertial terms cancel. There is also a small tangency discrepancy because the motion derivatives and profile normals are evaluated from a finite interpolant. Neither effect calls for a fixed allowance in newtons. Measure the discrepancy at the actual pose instead.

Let $H = [\boldsymbol{n}_\ell \ \boldsymbol{n}_u]$ and $\boldsymbol{v}_x = (r', z')^T$. Exact closed-contact motion gives $\boldsymbol{v}_x^T \boldsymbol{n}_\ell = 0$ and $\boldsymbol{v}_x^T \boldsymbol{n}_u = n_{u,z}$: the fixed track does no work and the moving track translates with unit axial speed per unit $\dot{x}$. Define the two computed residuals by

$$(d_\ell, d_u) = \boldsymbol{v}_x^T H - (0, n_{u,z}), \qquad E = |d_\ell N_\ell| + |d_u N_u|. \tag{8.1}$$

Projecting equation (5.2) onto $\boldsymbol{v}_x$ shows that $|F_x + N_u n_{u,z}| \le E$, where F_x is the common boundary's force from equation (4.3). Projection onto $\boldsymbol{v}_x - (0,1)^T$ gives the corresponding fixed-track identity. Thus $E/|n_{i,z}|$ measures each recovered normal's discrepancy from its scalar virtual-work identity.

For arithmetic resolution, split the right side of equation (5.2) into its acceleration, curvature and centrifugal vectors $\boldsymbol{b}_k$ and set

$$S_i = \left[|H^{-1}| \sum_k |\boldsymbol{b}_k| \right]_i, \qquad \eta_i = \frac{E}{|n_{i,z}|} + 4096\,\epsilon \max(N_0, S_i). \tag{8.2}$$

Absolute values of vectors and matrices are componentwise. Here ϵ is machine precision and $N_0 = 1\,\mathrm{N}$ is the reference scale in CINDER's existing roundoff convention. The inverse matrix accounts for amplification by the contact directions. Only the reported admissibility margin is snapped to zero when $|N_i| \le \eta_i$; the raw reaction, guide wrench and common force law are unchanged. A normal below $-\eta_i$ is rejected.

The same calculation applies to section 7, using its momentum-jump vector in place of $\sum_k \boldsymbol{b}_k$, normal impulses in equation (8.1), and a $1\,\mathrm{N\,s}$ reference scale. These checks quantify local numerical consistency. They neither certify hardware capacity nor replace the compiler's separate interpolation diagnostics.

Limit	Implemented response
Invalid inputs	Reject non-finite values, impossible tensors, nonpositive thickness, invalid domains/count, and a nonzero tangential COM offset.
Geometric branch	Search all finite-face minima. Establish regular reachable motion and support smooth contact-set changes. Report unresolved enclosures, ambiguous assembly, insufficient support and nonsmooth motion separately. Flat common-normal contact sets retain clamping and guide-moment bounds.
Operating state	Evaluate both normal forces with the solved local acceleration; reject contact loss. Report the complete ideal-guide demand separately.
Velocity reset	Resolve both normal impulses and reject a selected reset requiring tension. Direct projection callers must check the returned margins.
Hardware envelope	Speed, stress, pressure, real guide capacity, supporting-solid clearance, and three-dimensional interference require additional supplied data and checks; no universal safe RPM is derived from these profiles.

The mechanism may mount on either pulley when both contact tracks co-rotate with its shaft. A relative sheave helix is rejected. The host retains the full local-to-shared acceleration relation and the separate sheave/cup mass ownership. Point/segment authoring and the frontend preview service remain a separate application change; no template cup radius, backing thickness or arbitrary sample dimension bound is introduced by this core contact representation. The research viewer now imports CINDER's Python mechanics and contracts exported response coefficients in JavaScript; it does not maintain another contact-force equation.

9 Verification and what it establishes

The straight case recovers the force form used in Messick's straight-ramp model [4]: $M_f' = 0$ and the steady closing force is $Mrr'\omega^2$, with total M replacing a per-weight mass times count. This tests the force scale and geometry convention, not the dimensions of a particular machine.

The core smoke tests also differentiate inertial Cartesian trajectories of eight material masses. Only direct geometric positions are reused; the expected force, shaft torque and kinetic energy do not use the four-map formula. Three profile families, forward/reverse/zero shaft rotation, and both host shaft channels are exercised. Other checks cover derivative consistency, COM changes at fixed geometry, opposite products of inertia at identical four maps, saved-document/schema round trips, named running contacts, and valid and invalid stop projections. A short coupled neutral-shift transient checks primary angular-momentum conservation while axial motion changes shaft inertia.

The existing fixed-pivot launch regression remains a separate check of the common-boundary migration. None of these checks measures real CVTech guide capacity, contact pressure, wear, or post-release motion. Their practical conclusion is that the stated constrained model supplies consistent continuous and impulsive mechanics and explicitly identifies conditions it cannot retain.

A Contact-coordinate derivatives for the interpolant

The compiler also needs derivatives of the material contact coordinate. With $h = u_R = f''/(b'' - f'')$ at an interior tangent, another differentiation gives

$$u_{RR} = \frac{f'''(1+h)^2 - b'''h^2}{b'' - f''}, \qquad u' = hR', \qquad u'' = hR'' + u_{RR}(R')^2. \tag{A.1}$$

All of these derivatives are zero for a retained material endpoint. Lines and quadratics have zero third derivative. For the circle used in section 2, with $d = \rho - \rho_0$,

$$f'' = \frac{\sigma a^2}{(a^2 - d^2)^{3/2}}, \qquad f''' = \frac{3\sigma a^2 d}{(a^2 - d^2)^{5/2}}. \tag{A.2}$$

The open-semicircle domain and branch checks keep these values finite. A contact-mode change requires a separate branch; this derivative construction does not assign a smoothed corner motion.

References

[1] *General Centrifugal Mechanisms: From Compatible Motion to Clamping Force and Shaft Coupling.* Revision 4; canonical CINDER 1.2.1 edition. Common-boundary document.

[2] *Clamping Mechanism Implementations: The Fixed-Pivot Roller Flyweight.* CINDER 1.2.0 edition. Fixed-pivot document.

[3] Kai Arseneau and W. Spencer Smith. *A First-Principles Dynamic Model of a Rubber V-Belt CVT for General Transient Simulation: With Mechanical Actuation and Traction-Limited Torque Transfer.* Supplied manuscript, October 1, 2026; Sections 3.2, 3.3 and 3.6.

[4] M. J. Messick. Supplied 2018 thesis, Section 9.4.1, printed page 109 (PDF page 118), Eq. 9.2. Used for the straight-ramp force limit; the dimensions in this note are synthetic.